

RESEARCH ARTICLE

Demographic mechanisms of snowshoe hare population cycles in Yukon, Canada

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Handling Editor: Marlène Gamelon**Abstract**

1. One hundred years have elapsed since Charles Elton (1924) described the periodic fluctuations in North American snowshoe hare abundance, yet mechanisms underlying 9–11-year population cycles in snowshoe hares continue to be debated.
2. We applied multistate capture–mark–recapture models to long-term field data (1977–2020) based on >20,000 captures of >7000 unique snowshoe hares (*Lepus americanus*) from Kluane Lake, Yukon, Canada, to estimate and model state-specific demographic parameters. Juveniles had the lowest and reproductive adult females the highest apparent survival. Apparent survival of all sex-age classes was highest during the mid- and late-breeding seasons and was generally better during the increase phase.
3. Conditional probability of females transitioning from non-reproductive to reproductive state, and reproductive females remaining in the reproductive state, increased substantially as the population transitioned from low to increase phase throughout the breeding season.
4. Analysis of stage-structured matrix population models revealed that population-dynamic characteristics were strongly phase-specific, and also varied across seasons, with the increase phases being characterized by high monthly asymptotic population growth rate. Snowshoe hares experienced short stage-specific generation time during the early breeding season across all phases; they experienced relatively long generation time during the increase and low phase of the mid-breeding season, and the increase and peak phase of the late breeding season.
5. Elasticity analyses showed that asymptotic population growth rate was proportionately most sensitive to changes in survival of adult females across all phases and seasons. However, retrospective life table response experiment analysis showed that rapid growth of the snowshoe hare populations during the increase phase was due to improvements in reproductive transitions and pre-weaning

survival, whereas population declines are caused primarily by reduced survival (primarily, pre-weaning survival), with reduced reproductive transitions and smaller litter sizes playing a secondary role.

6. Our results suggest that cyclic populations of snowshoe hares are characterized by complex demographic and population-dynamic patterns, depending on phase of the cycle and reproductive season, and that different demographic mechanisms underlie rapid population growth during the increase phase, and swift population declines as the population transitions from the peak to the decline phase. Because our study represents the first comprehensive demographic and population-dynamic study of a cyclic population, similar studies would be needed to test the generalities of our conclusions. Whereas density-dependent predation has been shown to be the primary cause of phase-related changes in survival, future research should focus on identifying mechanisms underlying phase-related changes in reproductive parameters.

KEYWORDS

age- and sex-specific survival, breeding probabilities, generation time, *Lepus americanus*, life table response experiment (LTRE) analysis, population cycles, sensitivities and elasticities, wildlife's 10-year cycle

1 | INTRODUCTION

The 'balance of nature', an ecological worldview subscribed to by Charles Darwin and other early (and some modern) ecologists, postulates that ecological systems are in a state of balance or equilibrium, and that if perturbed, the systems tend to return to their equilibrium states via negative feedback mechanisms (reviewed in Simberloff, 2014; Justus, 2021). This idea was called into question when Charles Elton (1924) reported that some mammalian populations in the arctic and subarctic regions of Europe and North America exhibit periodic fluctuations in numbers (hereafter, 'population cycles'), and that these fluctuations have a large impact on the reproductive success and abundance of sympatric mammalian and avian predators. Elton's foundational work and subsequent research revealed that snowshoe hares (*Lepus americanus*) in North American boreal forests exhibit large-scale, cyclic fluctuation in numbers with a periodicity of 9–11 years, that the amplitude of cycles is highly variable both among populations and within a single population from one cycle to the next. Elton (1924) also recognized that population cycles are synchronous and could behave like travelling waves on large spatial scales (Keith, 1990; Krebs et al., 2014). Although the causal mechanisms responsible for these cycles are yet to be fully explained, experimental studies have identified predators, quality and availability of food resources during winter, and maternally inherited, predator-caused stress and ensuing epigenetic changes as potentially important drivers (Boonstra, 2024; Boonstra, Hik, et al., 1998; Boonstra, Krebs, & Stenseth, 1998; Edwards et al., 2021; Keith, 1990; Krebs et al., 2018; Sheriff et al., 2015).

We concentrate here on demographic variables that are responsible for causing population cycles. Changes in abundance are a consequence of changes in demographic variables, but the relative contribution of vital demographic rates may vary as the population growth rate changes from one phase of the cycle to another (Caswell, 2001; Krebs, 2013; Oli, 2019; Oli & Dobson, 1999). Thus, discerning demographic changes that cause large-scale, phase-related changes in abundance is essential for a complete understanding of population cycles. A 15-year study of a cyclic snowshoe hare population in central Alberta, Canada (Cary & Keith, 1978; Keith & Windberg, 1978) revealed that the increase phase of the cycle was characterized by high survival rates, particularly of juvenile hares, that natality began to decline at the peak phase, declining further during the decline phase and recovering before the population entered the low phase. Recruitment rates also showed phase-related changes probably due to changes in the length of the breeding season and changes in third and fourth litter pregnancy rates. The most comprehensive demographic understanding that currently exists is based on a series of collaborative studies of a cyclic snowshoe hare population in Yukon, Canada (Krebs et al., 1986, 1995, 2018; Krebs, Boutin, & Boonstra, 2001; Sinclair et al., 1988). These studies reported cyclic changes in reproductive rates and highlighted the potentially important role of recruitment in driving population cycles (Hodges et al., 2001; Keith & Windberg, 1978; Krebs et al., 1986; Krebs, Boonstra, et al., 2001; Krebs, Boutin, & Boonstra, 2001; Oli et al., 2020). Earlier studies in central Alberta (the southern edge of the boreal forest) suggested that the third and fourth litter pregnancy rates play an important role in triggering population growth (Keith, 1990; Keith & Windberg, 1978), but

recent evidence from Yukon (the heart of the boreal forest) suggests that these litters may contribute little to snowshoe hare population dynamics because overwinter survival of the late litters is always low (Peers et al., 2021).

Despite a long history of snowshoe hare population cycle research, demographic changes responsible for large-scale, phase-related changes in abundance and the relative contribution of demographic parameters to phase-related changes in population growth rate remain unknown. Here, we apply multistate capture-mark-recapture (CMR) models (Hestbeck, 1991; Lebreton et al., 2009; Williams et al., 2002) to long-term (1977–2020) field data spanning five population cycles of snowshoe hares at the Kluane region of south-western Yukon, Canada, to estimate and model age- and sex-specific apparent survival (hereafter referred to as 'survival') rates, age- and reproductive state-specific female breeding probabilities. Using empirical estimates of survival, transition and reproductive rates, we constructed and analysed phase-specific matrix population models to calculate important population-dynamic parameters (e.g. finite asymptotic population growth rate, generation time), and performed prospective and retrospective perturbation analyses (Caswell, 2000, 2001) to discern the potential influence of different demographic parameters, and the actual contribution of specific vital demographic rates to phase-related changes in finite population growth rate as the population changes from one phase of the cycle to another. Based on our prior knowledge of snowshoe hare ecology and life-history theory, we test the following predictions: (1) adults (≥ 1 -year old) would survive better than juveniles (young of the year) (Krebs et al., 1986) during all cyclic phases; (2) reproductive adult females would experience a lower survival than non-reproductive adult females due to the cost of reproduction associated with pregnancy and lactation (Stearns, 1989); (3) survival of all sex- and age classes would exhibit seasonal patterns with lower survival during winter than during summer months, and phase-specific patterns with lower survival during decline compared with increase and peak phases (Keith & Windberg, 1978; Krebs et al., 1986); and (4) the probability that non-reproductive females become reproductive, and reproductive females remain reproductive would be higher during the increase/peak phase than during the decline phase, and would be higher during early (EB) than late breeding (LB) season. Because demographic parameters in cyclic snowshoe hare populations vary across phases, we further expected (5) population-dynamic parameters (e.g. finite population growth rate, generation time) as well as the pattern of sensitivities and elasticities to vary across phases; however, based on earlier studies of sensitivity and elasticity patterns in birds and mammals (Gaillard et al., 2005; Oli & Dobson, 2003, 2005; Stahl & Oli, 2006), we also expected population growth rate to be more sensitive to changes in survival rates when the population is in the decline phase, and to reproductive parameters when the population is experiencing rapid growth. Finally, the demographic buffering hypothesis postulates that natural selection should reduce temporal variability in vital rates to which the population growth rate is most sensitive (Hilde et al., 2020); thus, we expected (6) vital rates to which the population growth rate is highly

sensitive would show less variation across phases and thus would not always make the largest contributions to phase-related changes in population growth rate.

2 | MATERIALS AND METHODS

2.1 | Study area and study species

Our field study was conducted near Lhù'ààn Mân' (Kluane Lake) in the south-western Yukon, Canada (details in Krebs, Boonstra, et al. (2001) and Krebs, Boutin, and Boonstra (2001)). Vegetation in this region is boreal forest dominated by white spruce (*Picea glauca*), with small areas of balsam poplar (*Populus balsamifera*) and trembling aspen (*Populus tremuloides*). Climate is cold, with mean monthly temperature falling below -20°C during winter months. The study area is typically under snow cover for about 8 months from October to May (Boonstra et al., 2016). The boreal ecosystem in the Kluane Lake region is largely undisturbed by human activities but has been changing over the last five decades due to global warming, a spruce bark beetle infestation and forest succession, and their effects on plant and animal communities (Krebs et al., 2019; Lavergne et al., 2021). A detailed description of the study area is given by Krebs, Boonstra, et al. (2001) and Krebs, Boutin, and Boonstra (2001).

Snowshoe hares are distributed across North American boreal forests (Hoffman & Smith, 2005), where they are one of the most important herbivores in terms of biomass and impact across trophic levels (Boonstra et al., 2016; Boutin et al., 1995; Krebs et al., 2018; Krebs, Boutin, & Boonstra, 2001). Their coat colour is pure white in winter but rusty brown during summer. Average lifespan is about 1 year but they can live up to 7 years of age (Boutin & Krebs, 1986; Hodges et al., 2001; Keith & Windberg, 1978; Krebs et al., 2018). Predation is the most important cause of hare mortality, with $\geq 90\%$ of hares dying in their youth due to predation (Hodges et al., 2001; Krebs et al., 2018). In our study system, Canada lynx (*Lynx canadensis*) and coyote (*Canis latrans*) are the most important predators, but most mammalian (e.g. red fox *Vulpes vulpes*, grey wolf *Canis lupus*), and avian (e.g. great-horned owl *Bubo virginianus*, goshawk *Accipiter gentilis*) predators of the boreal forest feed on snowshoe hares (Boutin & Krebs, 1986; Krebs, Boutin, & Boonstra, 2001). During summer, hares consume forbs, grasses, leaves and some woody browse, whereas the winter diet consists primarily of small twigs of dwarf birch (*Betula glandulosa*) willow (*Salix* spp.), and white spruce (*P. glauca*) (Hodges, 2000; Krebs et al., 2018; Smith et al., 1988).

Snowshoe hares at Kluane reproduce from May to September, with females producing up to 3 or 4 litters per breeding season. There is no evidence from Kluane that juvenile hares reproduce in the year of their birth (O'Donoghue & Krebs, 1992; Stefan & Krebs, 2001). Gestation lasts from 36 to 39 days, and parturition is synchronized so as to minimize early juvenile mortality due to predation (O'Donoghue & Boutin, 1995). Females can experience postpartum oestrus soon after parturition and mate again; thus, they are capable of both lactation and gestation simultaneously

(Naughton, 2012). Mean parturition dates for the first, second and third litters in Kluane mostly range from the third to the fourth week of May, last week of June to the first week of July and last week of July to the first week of August, respectively. Inter-litter intervals range from 27 to 38 days, but inter-litter intervals of <35 days are rare (Stefan & Krebs, 2001). In Kluane, mean litter size varies across litter groups, with the first litters being the smallest (3.6 leverets) and the second litters the largest (5.9–6.2 leverets) (Hodges et al., 2001). However, mean litter size can vary regionally and also depending on the phase of the population cycle (Cary & Keith, 1978; Hodges, 2000; O'Donoghue & Krebs, 1992; Stefan & Krebs, 2001). Early litters are weaned at 24–28 days of age, but late litters can be nursed for up to 40 days (O'Donoghue & Bergman, 1992). Juveniles of both sexes disperse predominantly between 30 and 60 days of age; leverets typically disperse by the end of September (Boutin et al., 1985; Gillis & Krebs, 1999; Hodges et al., 2001). Reproductive parameters show cyclic variation (Cary & Keith, 1978; Hodges et al., 2001; O'Donoghue & Krebs, 1992), but there is no evidence that dispersal rate varies depending on the phase of the population cycle (Boutin et al., 1985).

2.2 | Field methods

From 1977 to 2020, we live-captured snowshoe hares using Tomahawk live traps (Tomahawk Live Trap Co., Tomahawk, Wisconsin) located on 10m×10m or 20m×20m grids with 30m spacing between traps. Each hare received a Monel #3 eartag (National Band and Tag Co., Newport Kentucky) at first capture; the identity of previously marked hares was recorded in subsequent captures. Sex and reproductive status of adult females was determined based on evidence of lactation. Live trapping began in May 1977 and continued until May 2020; each calendar month of capture was considered to be the sampling occasion; multiple captures for a hare within a month, when they occurred, were considered to be a single encounter for that sampling occasion. All of our data came from unmanipulated (control) grids; we did not use data from grids that received experimental treatments (e.g. Krebs et al., 2018; Krebs, Boutin, & Boonstra, 2001). Until 1996, hares were trapped in all months; trapping primarily occurred between April and October thereafter. Thus, demographic information for the non-breeding (NB) season is heavily influenced by data collected between 1977 and 1996. Hares with unknown sex, and adult females for which reproductive status could not be ascertained were excluded from our analyses. Field methods are described in detail elsewhere (Hodges et al., 2001, 2006; Krebs et al., 1986). Our study spanned five population cycles (cycle that peaked in 1980 = Cycle 1 and the recent cycle that peaked in 2017 = Cycle 5).

Live-trapping protocols were approved by the Government of Yukon Scientists and Explorers Act Licenses and Wildlife Permits, the University of British Columbia Animal Care Committee, the University of Alberta Animal Use and Care Committee, and the University of Toronto Animal Care Committee. Our field research

protocol followed the guidelines of the American Society of Mammalogists (Sikes, 2016).

2.3 | CMR analysis

We used a multistate CMR modelling framework (e.g. Lebreton et al., 2009; Nichols, 1992; Williams et al., 2002) for estimating and modelling sex- and age-specific survival, and reproductive transition probabilities, and to test hypotheses regarding phase-specific and seasonal variation in the aforementioned parameters. At each encounter, a snowshoe hare was placed in one of five 'states' based on age, sex and reproductive status. Hares were considered juveniles in the year of birth (state *m*; male young of the year; state *f*; female young of the year), where the year of birth is the period between birth and the following spring (9–12-month period, depending on when the leverets were born). They were considered adult if they survived to the following spring (≥9 months of age, depending on the litter order in which they were born); we divided adult hares into three states depending on sex and reproductive status: adult males (state *M*; ≥9 months of age), reproductive adult females (*R*; females; ≥9 months of age with external evidence of lactation) and non-reproductive adult females (*N*; females ≥9 months of age with no external evidence of lactation). Juvenile males that survive the year of birth would be encountered only as adult males in the following years; juvenile females would be encountered either as reproductive or non-reproductive adult females after the year of birth. Adult males would remain in the same state throughout their lives, but adult females could transition between reproductive and non-reproductive states during the breeding season (hereafter, 'reproductive transition').

We used multistate CMR models to estimate state-specific apparent survival (S^k ; the probability that a hare in state *k* survives from sampling occasion *t* to *t* + 1 and remains in the study area), capture probability (p^k ; the probability that a hare in state *k* is recaptured if it was alive and present in the study area) and transition probability (ψ^{jk} ; the probability that a hare in state *j* at sampling occasion *t* transitioned to state *k* at sampling occasion *t* + 1, conditional on survival). We allowed S^k to be affected by singular, additive and interactive effects of the phase of the cycle (increase, peak, decline and low), and reproductive season (EB season: May and June; mid-breeding season [MB]: July and August; LB season: September and October; and NB season: November to April). Because snowshoe hares do not reproduce during the NB season, the probability of transition in the NB season from all states to the reproductive adult state was fixed to zero. Likewise, biologically impossible transitions (e.g. transition from adult to juvenile states, males to reproductive or non-reproductive adult female states and vice versa) were also fixed to zero; we allowed biologically realistic transitions to occur from each state in time *t* to all other states in time *t* + 1 without any constraints. As there are no ecological differences (i.e. no sexual dimorphism) between juvenile males and females (young of the year), we allowed capture

and survival probabilities to be the same; all other states were permitted to have state-specific survival. We then modelled ψ and S as singular, additive and interactive (two-way only) functions of cyclic phases and reproductive season. To keep the number of models to a reasonable level, we used a sequential model-fitting approach using the *p*-first strategy (Morin et al., 2020). First, we allowed p^k to be affected by the singular, additive and interactive effects of the phase of the cycle, reproductive season and cycle number, while allowing S and ψ to be affected by the singular effect of phase, season and an additive effect of the two. Results of these analyses revealed that the best model structure for p^k was a three-way interaction effect of phase, season and cycle number (all three covariates treated as factor variables). However, to permit some level of flexibility in modelling p^k , we used the top four model structures for p^k , and allowed S and ψ to be affected by the singular, additive and interactive effects of phase and season. This approach allowed estimation and modelling of the parameters of interest (S and ψ) without any restriction, while keeping the number of models to a reasonable level.

Following Keith (1990), we defined cyclic phases based on the rate of annual (Spring to Spring) change in snowshoe hare density (R); a given year was classified to be in a decline phase if $R < 0.44$, and an increase phase if $R > 1.89$. All years between the decline and increase phases were defined as the low phase, and years between increase and the next decline were defined as the peak phase (Oli et al., 2020). Snowshoe hare densities were estimated from ≥ 3 days of live trapping during each sampling occasion within the spatially explicit capture-recapture modelling framework using program DENSITY (Efford et al., 2009; Oli et al., 2022). To account for missing data, we fixed $p = 0$ for sampling occasions when hares were not captured. In all analyses, we specified time intervals between successive sampling occasions in months; estimated survival probabilities thus represent monthly rates.

We implemented multistate CMR models using program MARK (White & Burnham, 1999) Version 10.1 implemented in the RMark package version 3.0 (Laake, 2013) for the R computing environment, version 4.3.1 (R Core Team 2024). The goodness-of-fit of the multistate CMR model using program UCARE (Choquet et al., 2009) provided evidence of slight over-dispersion ($\chi^2 = 2898.93$, $df = 1831$, $\hat{c} = 1.58$), so we performed quasi-likelihood adjustments (details in Table S1). We used an information-theoretic approach for model selection with quasi-likelihood adjusted Akaike's information criterion corrected for small sample size (QAIC_c) as a measure of model parsimony (Burnham & Anderson, 2002; Williams et al., 2002). The effect of aforementioned covariates was determined by comparing QAIC_c among models with and without covariates. Models with difference in QAIC_c ($\Delta QAIC_c$) ≤ 2 were considered to be equally plausible.

2.4 | Matrix population modelling

Using the state-specific estimates of demographic parameters, we constructed a female-only, stage-structured matrix population model

with three stages: juveniles (young of the year), non-reproductive adults (≥ 9 -month-old females, depending on the litter order in which they were born) with no evidence of reproduction, and reproductive adults (≥ 9 -months of age, depending on the litter order in which they were born) with evidence of reproduction. Because snowshoe hare demographic parameters vary across seasons and cyclic phases (Cary & Keith, 1978; Hodges et al., 2001; Keith & Windberg, 1978; Krebs et al., 1986), we constructed a post-breeding census matrix population model (Caswell, 2001) for each season and cyclic phase (i.e. one population projection matrix for each combination of season and cyclic phase). The population projection matrix $\mathbf{A}$ was of the form:

$$\mathbf{A} = \begin{bmatrix} s_1(1 - (\Psi_{fn} + \Psi_{fr})) & s_0 s_2 \Psi_{nr} l s / 2 & s_0 s_3 \Psi_{rr} l s / 2 \\ s_1 \Psi_{fn} & s_2(1 - \Psi_{nr}) & s_3(1 - \Psi_{rr}) \\ s_1 \Psi_{fr} & s_2 \Psi_{nr} & s_3 \Psi_{rr} \end{bmatrix},$$

where s_1 , s_2 and s_3 are monthly survival of juveniles, non-reproductive adult females and reproductive adult females, respectively; Ψ_{fn} and Ψ_{fr} represent the probability of juvenile females transitioning to become non-reproductive and reproductive female, respectively, conditional on survival; Ψ_{nr} is the probability of non-reproductive adult females transitioning to become reproductive adult females, conditional on survival; and Ψ_{rn} and Ψ_{rr} represent the probability of reproductive adult females transitioning to become non-reproductive adult females, and the probability of reproductive adult females remaining in the same state, respectively, conditional on survival. Finally, l represents the litter size at birth, and s_0 is the pre-weaning survival (i.e. survival from birth until weaning). For clarity, we note that pre-weaning survival s_0 is included in the fertility term because young hares do not enter the traps until after weaning, approximately at 3–4 weeks of age. We estimated s_1 , s_2 , s_3 , Ψ_{fn} , Ψ_{fr} , Ψ_{nr} , Ψ_{rn} and Ψ_{rr} from the multistate CMR analysis of long-term field data (1977–2020) as described previously. However, litter size and pre-weaning survival data were available for only one cycle (1989–1996); we used estimates of l and s_0 reported by earlier studies (Hodges et al., 2001; O'Donoghue & Krebs, 1992; Stefan, 1998; Stefan & Krebs, 2001). We assumed an even primary sex ratio. Because all vital demographic parameters are monthly rates, matrix $\mathbf{A}$ represents a monthly population projection matrix.

We calculated the season- and phase-specific monthly finite asymptotic population growth rate (λ) as the dominant eigenvalue of the matrix $\mathbf{A}$; stage-specific reproductive value (RV) was calculated as the left eigenvector corresponding to the dominant eigenvalue of matrix $\mathbf{A}$ scaled such that the RV of the juvenile stage is 1.0. Calculation of some demographic quantities for stage-structured matrix models (stage-specific RV, and life expectancy) necessitates decomposition of the matrix $\mathbf{A}$, such that $\mathbf{A} = \mathbf{T} + \mathbf{F}$, where $\mathbf{T}$ describes the transitions and $\mathbf{F}$ describes reproduction (Caswell, 2001, 110–111). The fundamental matrix is then defined as $\mathbf{N} = (\mathbf{I} - \mathbf{T})^{-1}$, where $\mathbf{I}$ is a 4×4 identity matrix, and $\mathbf{T}$ is the matrix of transition probabilities. Stage-specific life expectancies are then calculated as the column sum of the matrix $\mathbf{A}$ (equation 5.10; Caswell 2001). The net

reproductive rate R_0 is the dominant eigenvalue of the matrix $\mathbf{R} = \mathbf{F}\mathbf{N}$. Finally the generation time was calculated using equation 5.73 of Caswell (2001) as $T = \ln(R_0) / \ln(\lambda)$, where $\ln$ represents natural log.

We calculated the sensitivity of λ to changes in lower-level vital demographic parameter θ as the partial derivative of λ with respect to θ , and calculated proportional sensitivity (elasticity) as (Caswell, 2001; Morris & Doak, 2002):

$$e(\theta) = \frac{\partial \lambda}{\partial \theta} \frac{\theta}{\lambda},$$

where $\partial \lambda / \partial \theta$ is the sensitivity of λ to changes in a season- and phase-specific demographic parameter θ , and λ denotes the finite monthly population growth rate.

2.5 | Analysis of life table response experiments (LTRE)

Two important yet unanswered questions in population cycle research are: what demographic changes cause an increasing population to stop growing and then crash? And what demographic changes cause a small population in the low phase of the cycle to increase rapidly and achieve very high densities? We used retrospective LTRE analyses to address these questions (Caswell, 1989, 2001; Dobson et al., 2024). An LTRE analysis is a demographic technique that permits decomposition of observed differences (or changes) in population growth rates into its underlying contributions from vital demographic rates (Caswell, 2001; Divoky et al., 2023; Dobson et al., 2024; Oli et al., 2001). The difference in λ between a 'treatment' and a 'control' is decomposed as:

$$\lambda_{\text{treatment}} - \lambda_{\text{control}} = \Delta \lambda = \sum_i (\theta_{\text{treatment}} - \theta_{\text{control}}) \frac{\partial \lambda}{\partial \theta_i} \bigg|_{\frac{\theta_{\text{treatment}} + \theta_{\text{control}}}{2}}.$$

The sensitivities are evaluated at the mean parameter values (Caswell, 1989, 2001). Using this approach, we decomposed phase-specific differences in λ ($\Delta \lambda$) within each season. The contribution of a demographic parameter to the observed phase-specific differences in λ is the product of the differences in the parameter values and sensitivity of λ to that parameter evaluated at the mean value. LTRE contributions represent the relative contribution of different demographic parameters to differences in λ between populations characterized by these same phase- and season-specific vital rates.

Demographic analyses were performed by use of the *popbio* package (Stubben & Milligan, 2015) for R computing environment (R Core Team 2024). We note that all demographic quantities derived from matrix population models are asymptotic quantities, calculated under the assumption of stable age theory. For example, λ here represents the asymptotic monthly population growth rate under the assumption of stable or stage theory, and not the observed or realized population growth rate. Likewise, vital rate sensitivities, elasticities and LTRE contributions represent how λ would respond to small absolute (sensitivity) or proportional (elasticity) changes in vital rates, and the contribution of vital rates to changes in λ as

the population transitions from one phase of the cycle to another (Caswell, 2001).

3 | RESULTS

During 1977–2020, we captured 7224 snowshoe hares (3695 females; 3529 males) a total of 20,066 times. Adult males and non-reproductive females were captured most frequently, and reproductive females least frequently. We captured age-sex groups as follows: juvenile males: 19.86%, juvenile females: 16.41%, adult males: 30.27%, non-reproductive adult females: 24.92%, and reproductive adult females: 8.54%; $N = 20,066$. The largest number of juveniles were captured during the MB and LB season; the largest number of reproductive females were captured during the EB and MB season (when young of the year become trappable). Sampling was relatively sparse during the winter months, with very few hare captures (Appendix S1; Figure S1A).

The proportion of adult females that were reproductive (i.e. females with evidence of lactation) varied substantially across months as the breeding season progressed; reproduction began in May with about 23% of adult females being reproductive; this value peaked in July and August when about 70% of adult females were reproductive, and it then declined to 29% in September. However, the proportion of adult females that were reproductive varied substantially across cyclic phases, with a greater proportion of adult females being in reproductive state during the increase phases (Figure S1B).

3.1 | Survival and transition probabilities

Among sex and age classes, juveniles had the lowest (0.772 ± 0.005) and reproductive adult females had the highest (0.907 ± 0.009) monthly survival; adult males and non-reproductive adult females had similar survival (adult males: 0.879 ± 0.003 ; non-reproductive adult females: 0.844 ± 0.005) (Figure S2A). Considering the phase effect only, we found that state-specific survival was higher during the increase and peak, compared with the decline and low phases; this phase effect on monthly survival was particularly strong for juveniles (increase: 0.820 ± 0.009 ; decline: 0.729 ± 0.001) (Figure S3A). Transition probabilities also strongly exhibited phase-specific patterns, with the probability of transitioning from non-reproductive to reproductive adult female state being higher during the increase or peak, compared with the decline or low, phase (Figure S3B). Considered seasonally, survival was higher during the MB and LB season for both sexes and all age classes (Figure S4A). The probability that NB adult females become reproductive, and reproductive females remained reproductive (conditional on survival), was the highest during the MB season, followed by that in the EB season (Figure S3B). Comparing single covariate models, we found that a model that included the influence of cyclic phase on state-specific survival, and season on transition probabilities, was better supported than any other model ($\Delta\text{QAICc} = 12.46$; Table 1a). This model

TABLE 1 Model comparison statistics testing for the influence of cyclic phase (phase) and reproductive season (rep.season) on state-specific apparent survival (S), and state transition probabilities (ψ), and the influence of phase, reproductive season and cycle number on state-specific capture probabilities (p) of snowshoe hares in Kluane, Yukon, 1977–2020. Top 10 models are presented for analyses (a) testing for single covariate effects on S and ψ , and best model structures for p ; and (b) testing for singular, additive and interactive (two-way only) effects of phase and reproductive season on S and ψ .

Model no.	Model	K	QAICc	Δ QAICc	Weight	QDeviance
a						
1	$S(\text{state} * \text{phase})p(\text{stratum} * \text{phase} * \text{cycle_no} * \text{rep.season})\Psi(\text{state} * \text{rep.season})$	188	219,580.122	0.000	1.000	207,842.449
2	$S(\text{state} * \text{phase})p(\text{state} * \text{phase} * \text{cycle_no} * \text{rep.season})\Psi(\text{state} * \text{rep.season})$	156	219,603.120	22.998	0.000	207,930.787
3	$S(\text{state} * \text{phase})p(\text{state} * \text{phase} * \text{cycle_no} * \text{rep.season})\Psi(\text{state} + \text{rep.season})$	144	219,604.414	24.293	0.000	207,956.519
4	$S(\text{state} * \text{phase})p(\text{stratum} * \text{phase} * \text{cycle_no} * \text{rep.season})\Psi(\text{state} + \text{rep.season})$	176	219,607.148	27.027	0.000	207,894.008
5	$S(\text{state} + \text{phase})p(\text{state} * \text{phase} * \text{cycle_no} * \text{rep.season})\Psi(\text{state} * \text{rep.season})$	147	219,609.175	29.053	0.000	207,955.173
6	$S(\text{state} + \text{phase})p(\text{state} * \text{phase} * \text{cycle_no} * \text{rep.season})\Psi(\text{state} + \text{rep.season})$	135	219,613.496	33.374	0.000	207,983.905
7	$S(\text{state} + \text{phase})p(\text{stratum} * \text{phase} * \text{cycle_no} * \text{rep.season})\Psi(\text{state} + \text{rep.season})$	167	219,619.008	38.886	0.000	207,924.243
8	$S(\text{state} * \text{rep.season})p(\text{state} * \text{phase} * \text{cycle_no} * \text{rep.season})\Psi(\text{state} * \text{rep.season})$	156	219,626.016	45.894	0.000	207,953.683
9	$S(\text{state} + \text{phase})p(\sim \text{stratum} * \text{phase} * \text{cycle_no} * \text{rep.season})\Psi(\text{state} * \text{rep.season})$	179	219,631.727	51.606	0.000	207,912.457
10	$S(\text{state} + \text{rep.season})p(\text{state} * \text{phase} * \text{cycle_no} * \text{rep.season})\Psi(\text{state} * \text{rep.season})$	147	219,633.025	52.903	0.000	207,979.023
b						
1	$S(\text{state} + \text{rep.season} * \text{phase})p(\text{state} * \text{phase} * \text{cycle_no} * \text{rep.season})\Psi(\text{state} * \text{rep.season} + \text{phase})$	162	219,472.424	0.000	0.997	207,787.859
2	$S(\text{state} + \text{rep.season} * \text{phase})p(\text{state} * \text{phase} * \text{cycle_no} * \text{rep.season})\Psi(\text{state} * \text{rep.season})$	159	219,484.889	12.465	0.002	207,806.448
3	$S(\text{state} + \text{rep.season} * \text{phase})p(\text{stratum} * \text{phase} * \text{cycle_no} * \text{rep.season})\Psi(\text{state} * \text{rep.season} + \text{phase})$	194	219,486.031	13.607	0.001	207,736.086
4	$S(\text{state} + \text{rep.season} * \text{phase})p(\text{state} * \text{phase} * \text{cycle_no} * \text{rep.season})\Psi(\text{state} + \text{rep.season} + \text{phase})$	150	219,490.477	18.053	0.000	207,830.367
5	$S(\text{state} + \text{rep.season} * \text{phase})p(\text{stratum} * \text{phase} * \text{cycle_no} * \text{rep.season})\Psi(\text{state} * \text{rep.season})$	191	219,494.790	22.366	0.000	207,750.979
6	$S(\text{state} + \text{rep.season} * \text{phase})p(\text{state} * \text{phase} * \text{cycle_no} * \text{rep.season})\Psi(\text{state} + \text{rep.season} * \text{phase})$	159	219,495.431	23.007	0.000	207,816.983
7	$S(\text{state} + \text{rep.season} * \text{phase})p(\sim \text{stratum} * \text{phase} * \text{cycle_no} * \text{rep.season})\Psi(\text{state} + \text{rep.season} + \text{phase})$	182	219,505.593	33.169	0.000	207,780.191
8	$S(\text{state} * \text{rep.season} + \text{phase})p(\text{state} * \text{phase} * \text{cycle_no} * \text{rep.season})\Psi(\text{state} * \text{rep.season} + \text{phase})$	162	219,510.182	37.758	0.000	207,825.617
9	$S(\text{state} * \text{rep.season} * \text{phase})p(\text{state} * \text{phase} * \text{cycle_no} * \text{rep.season})\Psi(\text{state} * \text{rep.season})$	204	219,510.306	37.882	0.000	207,739.869
10	$S(\text{state} + \text{rep.season} + \text{phase})p(\text{state} * \text{phase} * \text{cycle_no} * \text{rep.season})\Psi(\text{state} * \text{rep.season} + \text{phase})$	153	219,510.710	38.286	0.000	207,844.490

Note: For each model, we report the number of parameters (K), quasi Akaike information criterion corrected for small sample size (QAICc), difference in QAICc (Δ QAICc) and model weight. A '+' indicates additive effects of the covariates, whereas an asterisk (*) indicates both additive and interactive effects. Seasons are: EB, early breeding season (May and June); MB, mid-breeding season (July and August); LB, late breeding season (September and October); and NB, non-breeding season (November to April). Cyclic phases are: D, decline; I, increase; L, low phase; P, peak. States are: J, juvenile males and females; M, adult males; N, non-reproductive adult females; R, reproductive adult females. Survival of juvenile males and juvenile females was constrained to be the same, and we allowed transition to occur from all states to all other biologically relevant states.

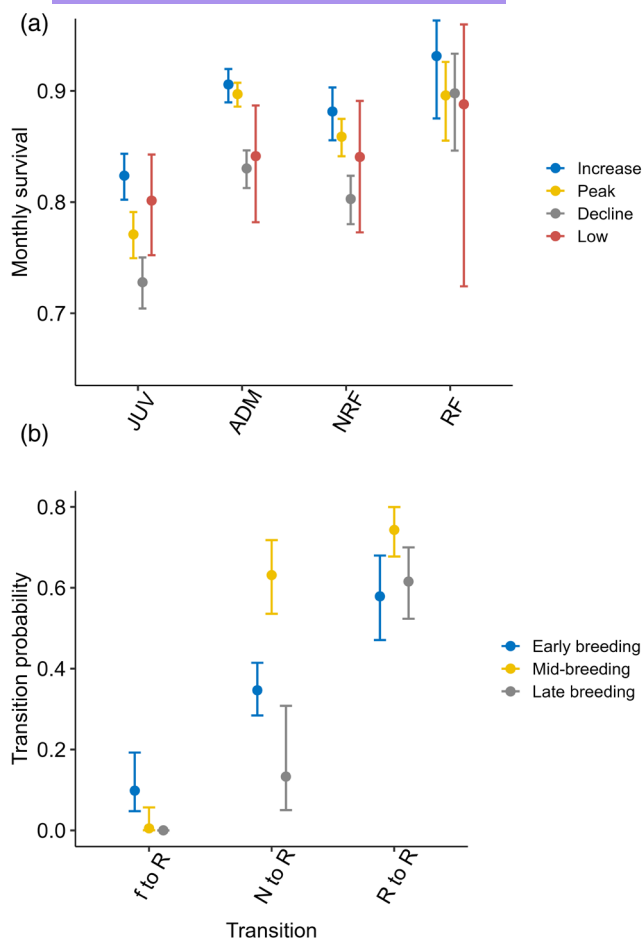


FIGURE 1 Estimates of survival (a) and transition probabilities (b) from the most parsimonious single covariate model (Model 1, Table 1a) for snowshoe hares in Kluane, Yukon, Canada, 1977–2020. States are: JUV, juvenile females and males; ADM, adult males; NRF, non-reproductive adult females and RF, reproductive adult females. Transition probabilities (conditional on survival) the symbols are: f to R, probability that a juvenile female transitions to reproductive adult; N to R, probability that a non-reproductive adult becomes reproductive adult and R to R, probability that reproductive females remain reproductive in subsequent encounters. All transition probabilities are conditional on survival. Reproductive seasons are: early breeding season (May and June), mid-breeding season (July and August), late breeding season (September), and non-breeding season (October to April). Cyclic phases are: I, increase; P, Peak; D, decline and L, low phase. Vertical bars represent 95% confidence intervals.

revealed a phase-specific pattern of survival and season-specific probability of transition as described above but also showed that season had a stronger influence on transition probabilities than cyclic phase (Figure 1a,b).

Phase- and season-specific variation in snowshoe hare demography was also reflected in the overall best model (Model 1, Table 1b), which revealed that cyclic phase and season interacted to influence sex- and age-specific survival, and transition probabilities, albeit slightly differently. Survival was higher for reproductive females than for other sex or age classes, and higher

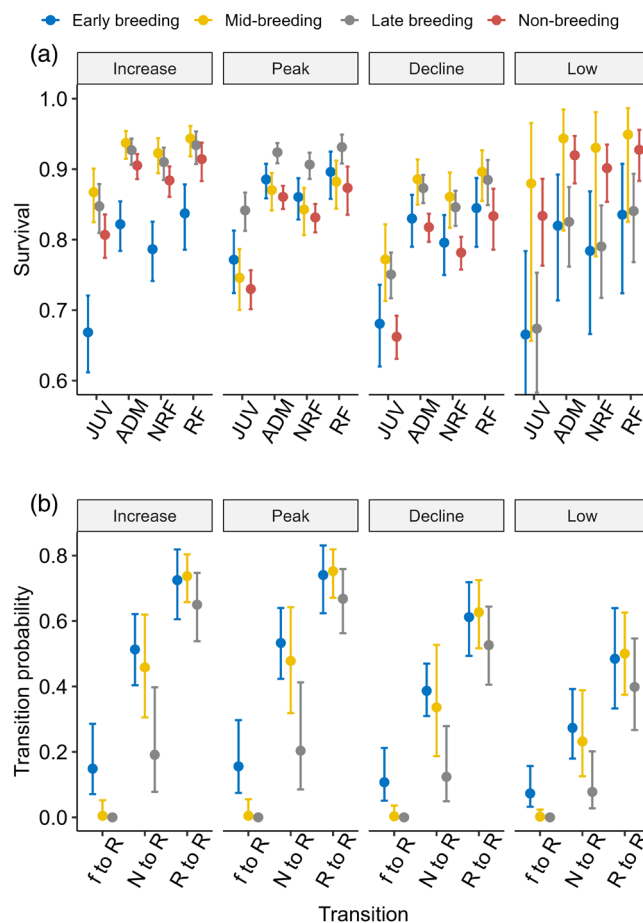


FIGURE 2 Estimates of survival (a) and transition probabilities (b) based on the most parsimonious overall model (Model 1, Table 1b) for snowshoe hares in Kluane, Yukon, Canada, 1977–2020. States are: JUV, juvenile males and females; ADM, adult males; NRF, non-reproductive adult females and RF, reproductive adult females. Transition probabilities (conditional on survival) are: f to R, probability that a juvenile female transitions to reproductive adult; N to R, probability that a non-reproductive adult becomes reproductive adult and R to R, probability that reproductive females remain reproductive in subsequent encounters. All transition probabilities are conditional on survival. Reproductive seasons are: early breeding season (May and June), mid-breeding season (July and August), late breeding season (September and October), and non-breeding season (November to April). Cyclic phases are: increase, peak, decline and low phase. Vertical bars represent 95% confidence intervals.

during MB and LB season. Survival of hares of both sexes and all age classes was highest during the increase phase across all seasons except the EB season; state-specific survival was similarly high in the increase and low phase during MB and NB seasons, but not during the EB and LB seasons (Figure 2a). The probability that non-reproductive adult females transitioned to become reproductive, and reproductive adult females remained reproductive (conditional on survival), was substantially higher during the increase and peak phases than the decline or low phases. The probability of transitioning from non-reproductive adult state to reproductive adult state, conditional on survival, declined as

the breeding season progressed (Figure 2b). Capture probability varied widely across states, phases, seasons and among cycles (Table 1; Figure S5).

3.2 | Pre-weaning survival and litter size

Pre-weaning survival was generally high during the increase phase across all seasons (≥ 0.57) and progressively declined as the population transitioned to the peak and decline phases. Pre-weaning survival was zero during the decline phase of MB season; no litters were produced during the decline phase in LB season (Figure S6A). The MB and LB seasons of the increase phase and MB season of the peak phase were characterized by the largest litters (≥ 5.8 leverets per litter); litter sizes were smaller during the EB season across all seasons (Figure S6B) (Hodges et al., 2001; O'Donoghue & Krebs, 1992; Stefan, 1998).

3.3 | Population growth rate, generation time, RVs and life expectancy

The monthly finite asymptotic population growth rate (λ) varied strongly across cyclic phases and reproductive seasons (Figure 3a). The increase phase was characterized by high asymptotic population growth rates throughout the breeding season, with the growth rate being highest during the MB season (monthly $\lambda = 1.286$; 95% CI = 1.201–1.370). During the peak phase, the population experienced rapid growth during the EB season but was generally stable during the MB and LB seasons. During the decline phase, λ was < 1.0 across all seasons, and during the low phase λ also was < 1.0 except during the MB season. During the NB season, the decline phase experienced the sharpest decline ($\lambda = 0.782$; 95% CI = 0.758–0.805; Figure 3a). Likewise, generation time also showed phase- and season-specific variation, with the hares in the EB season of the increase phase experiencing the shortest generation time (6.55 months), and those in the low phase of the MB season experiencing the longest generation time (15.87 months; Figure 3b).

RVs revealed that reproductive adult females were the most important stage in terms of their relative contribution to the population growth via reproduction (Figure 4a). However, the importance of reproductive (and non-reproductive) adult females was substantially higher during the LB season, but much lower during the NB and EB seasons. For example, RV of reproductive adults was 1.491 and 1.684 during the increase and peak phases of the NB season (relative to the RV of juveniles), respectively, whereas it was 7.554 and 4.483 during the low and increase phases of the LB season, respectively (Figure 4a). Stage-specific expectation of further life (life expectancy) also varied across breeding seasons and cyclic phases. Across all seasons and phases, adult females (especially reproductive females) had higher life expectancy; however, stage-specific differences in life expectancy were small during the EB season. Interestingly, hares in all stages had high life expectancies during the low and increase phases of the MB season; life expectancy was

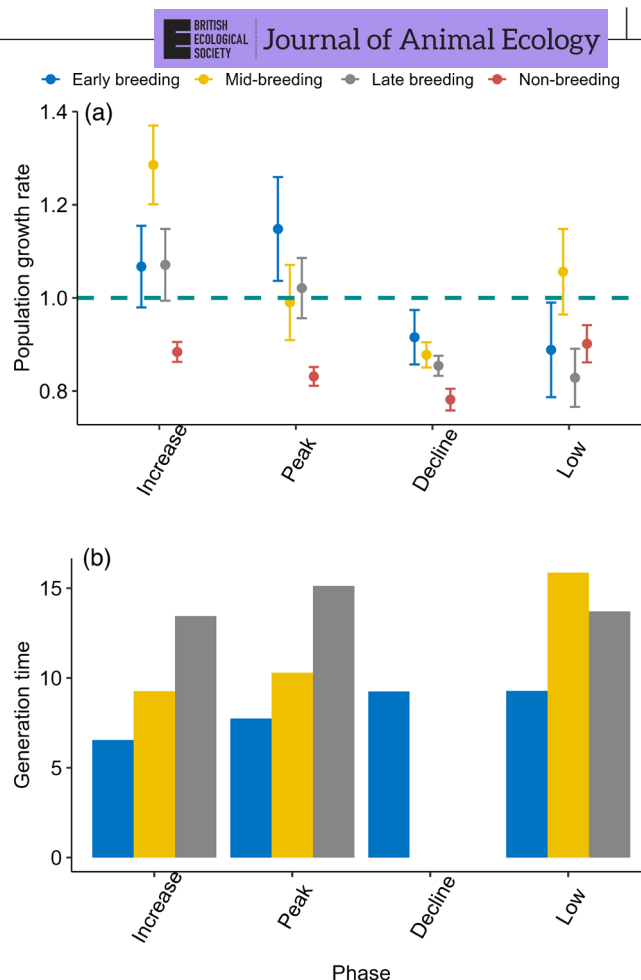


FIGURE 3 Reproductive season- and phase-specific estimates of monthly finite asymptotic population growth rate (λ) (a) and generation time (months) (b) for snowshoe hares in Kluane, Yukon, Canada. Reproductive seasons are: early breeding season (May and June), mid-breeding season (July and August), late breeding season (September–October) and non-breeding season (November and April). The horizontal dashed line represents zero population growth ($\lambda = 1$). Vertical lines in panel (a) represent 95% confidence intervals.

low for all stages; during the decline and low phases of the EB season and peak and decline phases of the LB season. For example, juveniles, non-reproductive and reproductive adult females in the increase phase were expected to survive an additional 12.6, 15.3 and 15.8 months, respectively; but in the decline phase they were expected to survive only 3.6, 4.9 and 5.3 additional months, respectively (Figure 4b).

3.4 | Prospective perturbation analyses

Prospective perturbation analyses revealed that sensitivity and elasticity of λ to changes in vital demographic parameters varied across reproductive seasons, and within each season, across cyclic phases (Figure 5a; Figure S7). Generally, however, λ was proportionately most sensitive to changes in survival of reproductive and non-reproductive adult females across all phases and seasons (Figure 5a).

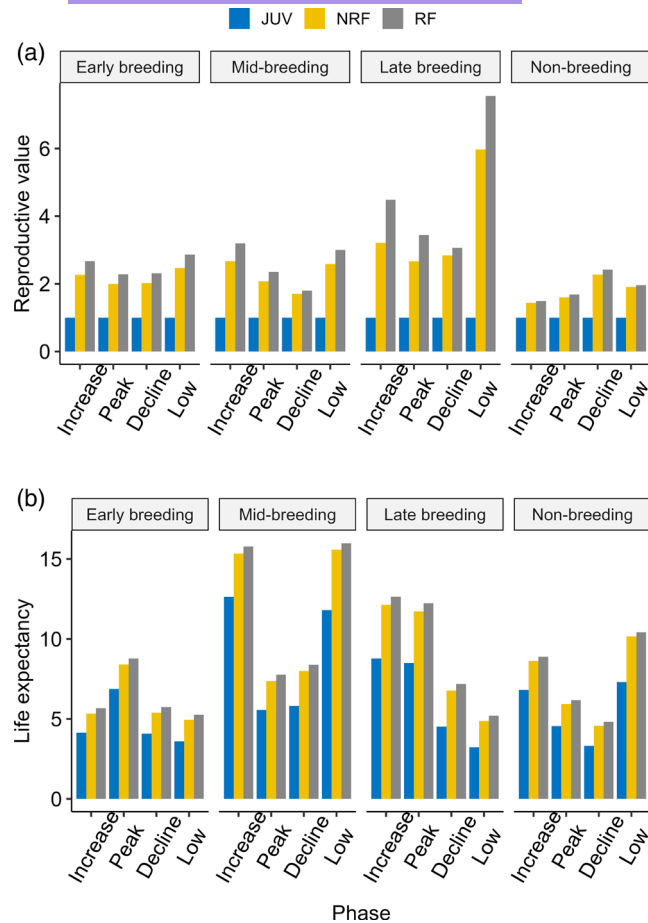


FIGURE 4 Stage-specific reproductive values (a) and life expectancy (b) for a snowshoe hare population in Kluane, Yukon, Canada, for each reproductive season and cyclic phase. Reproductive values for juveniles are set to 1.0; reproductive values for other stages are interpreted relative to that for juveniles. Reproductive seasons are: early breeding season (May and June), mid-breeding season (July and August), late breeding season (September–October) and non-breeding season (November and April). Stages are: JUV, juveniles; NRF, non-reproductive adult females and RF, reproductive adult females.

Sensitivities revealed a similar pattern, except that the importance of transition to reproductive state and pre-weaning survival became as important as or more important than adult survival during the MB and LB seasons (Figure S7). Both the sensitivity and elasticity of λ to changes in litter size were low across all seasons and phases, suggesting relatively small potential influence of this variable on population growth rate (Figure 5a; Figure S7). During the NB season, λ had the highest sensitivity and elasticity to changes in survival of non-reproductive adult females.

3.5 | Retrospective perturbation analyses: LTRE contribution to $\Delta\lambda$

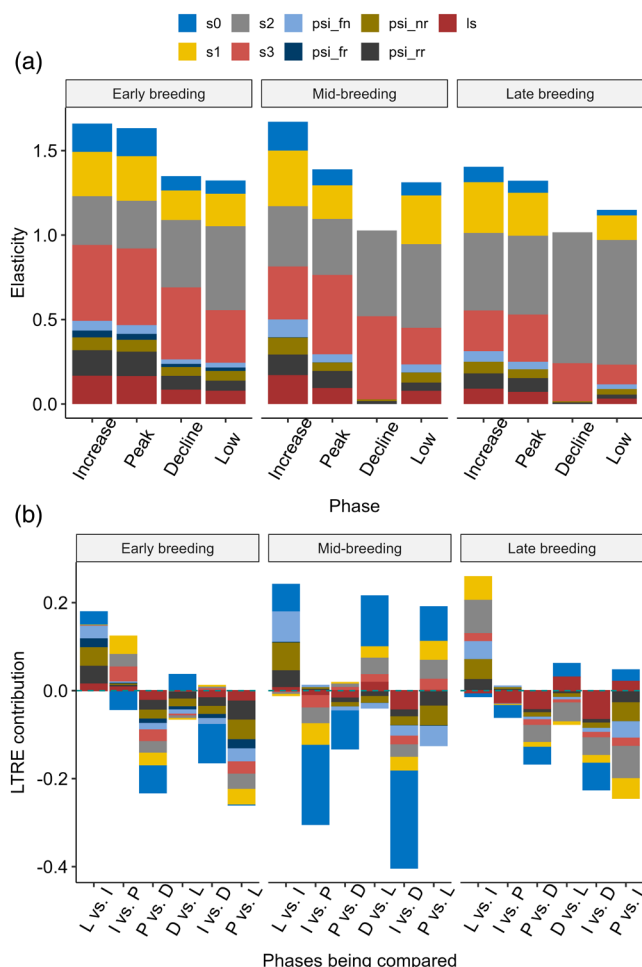
For each season, we calculated the phase-specific differences in population growth rate ($\Delta\lambda$; Figure S8) and then applied LTRE analysis to

decompose $\Delta\lambda$ into contributions from each vital demographic parameter (Figure 5b). Across all seasons except the NB season, the monthly population growth rate increased as the population transitioned from low to increase phase ($\Delta\lambda=0.179\text{--}0.243$; Figure S8). These increases in λ were primarily a result of improved transition to reproductive states, with improved pre-weaning survival and litter size playing a secondary role; survival of juveniles or adults made little contribution except during the LB season (Figure 5b). The population growth rate declined as the population moved from increase to peak (except in EB season) and from peak to decline; these declines in population growth rate occurred primarily due to a reduction in the survival of all age classes, with pre-weaning survival playing a more important role. Phase-specific differences in λ were particularly strong during the MB season with the increase phase being characterized by the very high population growth rate ($\lambda=1.285$), and the difference in λ between the increase and decline phase was quite drastic ($\Delta\lambda=-0.408$; Figure S8). Our results revealed that pre-weaning survival was a critically important parameter that contributed substantially to $\Delta\lambda$ across all phases and breeding seasons (Figure 5b). Across all seasons, declines in λ occurred primarily due to a reduction in survival, whereas increases in λ were primarily a consequence of improved pre-weaning survival and reproductive transition. During the NB season, λ marginally declined as the population transitioned from low to increase, increase to peak, and peak to decline phases ($\Delta\lambda=-0.017$ to -0.053); however, λ increased as the population moved from decline to low phase ($\Delta\lambda=0.120$). All phase-specific changes in λ during the NB season were due to changes in the survival of non-reproductive adult females; all females became non-reproductive in this season.

4 | DISCUSSION

4.1 | Seasonal and phase-specific pattern of snowshoe hare demographic parameters

Considered by some to be one of the most intriguing mysteries in ecology (Myers, 2018; Stenseth, 1999), population cycles in snowshoe hares and Arvicoline rodents in northern latitudes have fascinated generations of ecologists after they were first eloquently described by Elton (1924) a century ago. In our study, we aimed to advance our knowledge of snowshoe hare demography during population cycles based on a thorough CMR analysis of field data spanning five population cycles in the Kluane region, and construction and analysis of matrix population models to discern phase-related demographic and population-dynamic characteristics. As expected, monthly survival varied across sex- and age classes, with juvenile hares having the lowest survival rate, but contrary to our expectation based on life-history theory (Stearns, 1992), reproductive adult females experienced the highest survival, higher than that of non-reproductive adult females or adult males. One possible explanation for this result is that reproductive females may be more risk averse. Within each age class, monthly survival varied substantially across



cyclic phases with survival generally being the highest during the increase phase (juveniles: 82%; adult males: 90%; non-reproductive adult females: 88%; reproductive adults: 93%) and lowest during the decline phase (juveniles: 73%; adult males: 83%; non-reproductive adult females: 80%; and reproductive adult females: 89% per month). Phase-specific variation in survival of juveniles was particularly high and ranged from 73% in the decline phase to 82% during the increase phase. Assuming that juvenile females would have to survive on average 9 months from weaning until the first breeding opportunity, only about 6% ($0.73^9 = 0.06$) of them would survive to their first reproductive opportunity during the decline phase versus 17% ($0.82^9 = 0.17$), three times higher, during the increase phase. Peers et al. (2021) reported that overwinter survival of early litter juveniles was twice as high in the increase than in the decline phase, and Keith and Windberg (1978) noted that survival of juveniles from birth until mid-winter was about four times higher in the increase than in the decline phase. Taken together, these results suggest that survival of juveniles until they breed is an important driver of snowshoe hare population cycles. Phase-specific variation in survival of adult females was less drastic (~4%–5% per month).

Seasonally, state-specific survival was the lowest during the EB season (May to June), increased as the breeding season progressed, achieving the highest survival during the MB (July to August) or LB (September to October) season. Overwinter or NB season

FIGURE 5 (a) Elasticity of snowshoe hare finite population growth rate (λ) in Kluane, Yukon, Canada, to changes in pre-weaning survival (s_0), survival of juveniles (s_1), survival of non-reproductive adult females (s_2), survival of reproductive adult females (s_3), conditional transition probability of juveniles to become non-reproductive (ψ_{fn} ; Ψ_{fn}) and reproductive (ψ_{fr} ; Ψ_{fr}) adults, conditional transition probability of non-reproductive adult females to become reproductive (ψ_{nr} ; Ψ_{nr}), conditional transition probability of reproductive females to become non-reproductive (ψ_{rn} ; Ψ_{rn}) and conditional transition probability of reproductive female to remain reproductive (ψ_{rr} ; Ψ_{rr}). Elasticities are given for each cyclic phase (increase, peak, decline and low), and reproductive season (early breeding, mid-breeding and late breeding). Results are not shown for the non-breeding season because snowshoe hares do not reproduce during the winter non-breeding season; consequently, only state-specific survival rates are non-zero. (b) Life table response experiment (LTRE) contribution of vital rates to observed phase-specific differences in finite monthly population growth rate ($\Delta\lambda$) in Kluane, Yukon, Canada. Vital rates are: pre-weaning survival (s_0), survival of juveniles (s_1), survival of non-reproductive adult females (s_2), survival of reproductive adult females (s_3), conditional transition probability of juveniles to become non-reproductive (ψ_{fn} ; Ψ_{fn}) and reproductive (ψ_{fr} ; Ψ_{fr}) adults, conditional transition probability of non-reproductive adult females to become reproductive (ψ_{nr} ; Ψ_{nr}), conditional transition probability of reproductive females to become non-reproductive (ψ_{rn} ; Ψ_{rn}), and conditional transition probability of reproductive female to remain reproductive (ψ_{rr} ; Ψ_{rr}). LTRE contributions are given for each pair of cyclic phases being compared (increase [I], peak [P], decline [D] and low [L]), and for each reproductive season (early breeding, mid-breeding and late breeding season). Length of each bar represents the observed difference in λ ($\Delta\lambda$) between the phases being compared; each colour shows the contribution of a demographic variable to $\Delta\lambda$. Results are not shown for the non-breeding season because snowshoe hares do not reproduce during the winter non-breeding season; consequently, only state-specific survival rates are non-zero.

(November to April) survival of adult hares of both sexes declined to a level similar to that in the EB season; however, the monthly overwinter survival of juvenile hares remained relatively high. For example, EB season monthly juvenile survival was 68%; it rose to 80% during the LB season and remained high overwinter at 77%; in contrast, the survival of non-reproductive adult females during EB, MB, LB and NB season was 84%, 92%, 91% and 83%, respectively (Figure S3A). The high overwinter survival of juveniles likely reflects the fact that, by the end of the breeding season, all young of the year may have dispersed (Gillis & Krebs, 1999) and permanent emigration would be low, and surviving leverets born in first or second litters would have grown in size to be similar to adults, and that leverets born in the first or second litters survive better than those born in subsequent litters (Keith & Windberg, 1978; Peers et al., 2021).

The most parsimonious survival model revealed that both season and cyclic phase interacted to influence state-specific survival (Table 1b, Model no. 1). Although the phase- and season-specific pattern of survival described above generally held, it also revealed that: (1) for all sex and age classes, EB season survival was the highest during the peak; (2) during MB and NB seasons, survival was similarly high in increase and low phases, and lower during the peak

and decline phases; and (3) during the MB season, survival declined progressively from increase, peak, decline and low phase; this survival decline was particularly drastic for juveniles which declined from 85% in the increase phase to 67% in the low phase, a decline in monthly survival of 18%. With the exception of the EB season, peak or decline phases were generally characterized by low survival, and in some phase-season combinations, survival in low phase was similar to or better than that in the increase phase (Figure 3a), a finding also reported by Hodges et al. (1999). These results reveal a complex interaction of cyclic phase and reproductive season to influence age- and sex-specific survival and provide support for the potentially important role of juvenile survival in driving snowshoe hare population cycles (Keith & Windberg, 1978; Peers et al., 2021).

Phase specificity of snowshoe hare survival was first described by Keith and Windberg (1978) based on their 15-year study of a cyclic snowshoe hare population in central Alberta, Canada. They noted that juvenile survival from birth until the following winter (5–6 months) ranged from 6% to 11%, and that these rates were ~three to four times higher during years of population growth (increase phase) than during years of population decline (Keith & Windberg, 1978). Our results suggest that 6-month juvenile survival was ~14% during the decline phase and ~30% during the increase phase. Annual survival of adult hares in Alberta was 11% during the decline and 34% during the increase phase. Our annualized survival estimates for non-reproductive adult females during the decline and increase phases were 9% and 28%, respectively. Although phase- and age-specific survival patterns in our study were generally consistent with those reported by Keith and Windberg (1978), we point out that estimation methods differed between the two studies. In an earlier analysis using a subset of our data from the first cycle (1977–1984), Krebs et al. (1986) showed that overall snowshoe hare survival in Kluane decreased progressively from increase and peak to decline and low phases. Similar to the pattern of seasonal variation recorded in this study, Krebs et al. (1986) found that juvenile survival generally increased from summer and autumn to winter. Based on a decade-long radio-telemetry study conducted during the Kluane experiment (1986–1996), Hodges et al. (2001) reported monthly adult survival during the increase phase on control grids to be 91%, which compares favourably with our estimates (adult males: 90%, non-reproductive adult females: 88%, and reproductive adult females: 93%). During the decline phase, survival was substantially lower, at 64%, which was lower than what we found for adult hares (Hodges et al., 2001). One possible explanation for this difference is that our estimates are based on data collected across five population cycles, and potentially reflect estimates averaged across cycles. A recent study in Kluane also highlighted the potentially important role of first and second litter juvenile survival in driving snowshoe hare population cycles (Peers et al., 2021).

As expected from the strong seasonality in hare reproduction, the probability of females transitioning from non-reproductive to reproductive state exhibited a strongly seasonal pattern—virtually all single covariate models for transition probability included reproductive season as a covariate (Model no. 1, Table 1a). Conditional

on survival, the probability of females transitioning from non-reproductive to reproductive (N to R transition) was about 35% during the EB season, rose to 63% during the MB season, then declined to about 13% by the end of the breeding season. Once reproductive, females remained reproductive (R to R transition) with high probability throughout the breeding season (Figure S3B). However, the model that also included an additive effect of phase on reproductive transition probability was substantially better supported ($\Delta\text{QAICc}=22.99$), and represented the overall best model (Model no. 1, Table 1b). N to R transition probabilities were generally higher during the increase and peak phase than decline and low phase, especially during the EB and MB seasons. These probabilities were low across all phases during the LB season (Figure 3b). For example, phase-specific N to R transition probabilities during the EB season were 51%, 53%, 39% and 27% in increase, peak, decline and low phase, respectively. One remarkable result was that non-reproductive females were 24%–45% more likely to become reproductive during the increase or peak phase, compared with the decline phase. The same seasonal and phase-specific pattern generally held for reproductive females that remained reproductive, with transition probability being higher in increase and peak phase than decline and low phases; however, these probabilities did not substantially differ between EB and MB season. Across all phases and seasons, the likelihood of non-reproductive females transitioning to become reproductive females, and reproductive females remaining reproductive was substantially higher during the increase and peak phases than the decline and low phases. Reproductive transitions, as we measured them, involve two elements. The first is the probability that a female gives birth to a given litter but the second is how long that litter survives thereafter because this affects the amount of time a female could be trapped and recorded as lactating (reproductive). For example, a female may give birth to a litter but lose that litter soon after (O'Donoghue & Boutin, 1995) leading to a stoppage of lactation. The female would then not show evidence of lactation until the birth of the next litter (~3–4 weeks). Such instances would contribute to a reduction in the R to R transition probability. Thus, R to R and N to R transition probabilities are a function of both the probability of producing a litter and the survival of that litter to weaning.

Cary and Keith (1978) reported that most reproductive parameters of snowshoe hares in Alberta, Canada, showed cyclic changes, and that improvements in reproductive parameters preceded the density cycles by about 3 years. We found that the proportion of females that were reproductive was generally low in May, peaked during June–July, and then declined later in the breeding season, but this proportion varied strongly across phases (Figure S1B). Phase-specific variation in the observed proportion of females that were reproductive was the greatest for the first litter (May), ranging from 21% during the low phase to 36% during the increase phase (Figure S1B). Pre-weaning survival showed stronger phase- and season-specific variation; it ranged from 0% in the decline phase of MB season to 64% in the increase phase of EB season (Figure S6A). Average litter sizes also varied across the breeding

season and cyclic phases and ranged from 3.2 leverets in the low phase of EB season to 6.4 leverets during the low of the LB season. Litter sizes were smaller during the EB than MB or LB season (Figure S6B). Generally, pre-weaning survival showed proportionately greater seasonal and phase-specific variation than litter size or survival of other age classes. Additional details regarding reproductive output and pre-weaning survival of snowshoe hares in our study site are provided by Stefan (1998), Stefan and Krebs (2001), and Hodges et al. (2001).

4.2 | Population growth, generation time and sensitivity analyses

As expected from the strong seasonal and phase-specific survival and reproductive parameters, the population growth rate also varied strongly across seasons and among cyclic phases. Across all seasons, monthly $\lambda > 1$ during the increase and peak phase across all seasons except the NB season, with the peak phase of EB season ($\lambda = 1.155$ per month) and the increase phase of MB season ($\lambda = 1.286$ per month) being characterized by the highest growth rate; during the decline phase $\lambda < 1$ in all seasons except in MB season when the population experienced ~15% monthly increase (Figure 3a). Because snowshoe hares do not reproduce during the winter period, the hare population always declined during winter, but the rate of decline varied across cyclic phases, with the slowest decline during the increase and low phases (6%–10% monthly decline), and the steepest decline during the decline phase (~20% monthly decline). Our estimates of λ are generally consistent with broad phase-specific patterns reported by earlier studies (Hodges et al., 2001; Keith & Windberg, 1978; Krebs et al., 1986), although matrix population model-based estimates of λ have never been reported previously.

The generation time is often considered to be an indicator of the tempo of life history in birds and mammals, with short generation time being indicative of fast life history and long generation time being associated with the slow life histories (Gaillard et al., 2005; Oli, 2004; Stahl & Oli, 2006). In our study population, generation time varied strongly based on the phase of the cycle, and secondarily based on the reproductive seasons, ranging from ~7 months in the increase phase in the EB season to ~15 months in the peak and low phase in the LB season, a 2.4-fold difference. It appears that snowshoe hares have a fast life history as the population transitions from the low to the increase phase, but the pace of life is substantially slower during the peak or low phase; across seasons, the pace of life becomes slower and slower as the breeding season progresses (Figure 3b). Likewise, RVs and stage-specific life expectancy also exhibited strongly seasonal and phase-specific patterns (Figure 4). Across all seasons and phases reproductive females had the highest RV, suggesting that reproductive females make the largest contribution to the population growth rate via reproduction (Caswell, 2001). However, the relative contribution of reproductive females strongly varied across seasons and phases. For example, RV of reproductive females (relative to that of the juveniles) was 1.49 in the increase phase of the NB season; it

was 7.55 in the low phase of the LB season, suggesting that reproductive females are five times more important to the population growth when the population is during the low phase in the LB season. Stage-specific life expectancy (i.e. expectation of further life conditional on surviving to a given stage; Caswell, 2001) also varied among stages with juveniles and reproductive females having the shortest and longest life expectancy across seasons and phases. However, there was a strong seasonal and phase-specific pattern. For example, juvenile life expectancy (i.e. life expectancy at weaning) ranged from 3.223 months in the low phase of the LB season to 12.635 months in the increase phase of the MB season, nearly a four-fold difference; individuals of all stages had the highest life expectancy in the increase and low phases of the MB season (Figure 4b).

The sensitivity and elasticity of λ to changes in vital rates have received substantial attention in ecological literature (Heppell et al., 2000; Oli & Dobson, 2003; Stahl & Oli, 2006). Broadly, in species with 'slow' life histories (i.e. species characterized by late maturity, low reproductive output and long lifespan, Promislow & Harvey, 1990; Oli, 2004; Gaillard et al., 2005; Healy et al., 2019), λ is proportionately more sensitive to changes in survival rates; in species with 'fast' life histories (i.e. species characterized by early maturity, high reproductive output and short lifespan) λ is proportionately more sensitive to changes in reproductive parameters. Based on the phase-related demographic characteristics of cyclic snowshoe hare populations, we expected λ to be proportionately more sensitive to survival rates during the decline and low phase, and to reproductive parameters during the increase and peak phase. Although the elasticity of λ to the survival of adult females was generally the greatest across all seasons and phases, the elasticity of λ to reproductive parameters (i.e. transition to reproductive state, litter size and pre-weaning survival which is a component of recruitment) increased during the increase and peak phase. Interestingly, the elasticity of λ to adult survival dominated the elasticity pattern during the decline phase, especially during the MB and LB season (Figure 5a).

4.3 | LTRE analyses

The environmental canalization and demographic buffering hypotheses postulate that natural selection should minimize temporal variation in demographic parameters to which population growth rate is highly sensitive (Gaillard & Yoccoz, 2003; Hilde et al., 2020). Sensitivity analyses revealed that λ was generally most sensitive to changes in the survival of adult females, so we expected relatively small changes in these rates as the population moved from one phase of the cycle to another. Results of LTRE analyses suggest that patterns of variation and LTRE contributions were much more complex than those predicted by the demographic buffering hypothesis and depended strongly on phases being compared and the reproductive season. During the EB and MB seasons, there was little or no change in the survival of adult females as the population transitioned from low to increase phase. Consequently, these variables contributed little to the observed increases in λ ; in contrast, the transition to

reproductive state and pre-weaning survival made the largest contributions except during the LB season when juvenile and adult survival also made sizeable contributions. The observed decrease in λ between peak and decline, and increase and decline, phases was primarily due to reduced pre-weaning survival although reduced adult survival and litter size also made sizeable contributions. Greatest phase-related differences in λ occurred during the MB season, suggesting that phase-related changes in demographic parameters during the peak breeding season are critical for driving cyclic snowshoe hare population dynamics. Our LTRE results suggest that (i) population growth as the population transitions from low to increase phase is driven primarily by improvements in reproductive transitions, abetted secondarily by improvements in pre-weaning survival; (ii) population crashes occur primarily due to reduced pre-weaning survival; however, reductions in the survival of other age classes as well as reproductive parameters also played important roles; and that (iii) pre-weaning survival is a critically important parameter that plays a prominent role in driving snowshoe hare population dynamics. We note, however, that estimates of pre-weaning survival and litter size data used in our analyses come primarily from one cycle, whereas estimates of state-specific survival and transition probabilities are based on five cycles. Thus, the strong influence of changes in pre-weaning survival on phase-related changes in population growth rate should be interpreted with caution.

In this analysis, we focused on discerning broader phase-specific, seasonal demographic and population-dynamic patterns. However, we know that the hare's environment is changing over time; consequently, each snowshoe hare population cycle is somewhat different in terms of peak densities, cycle amplitude and duration of cyclic phases (Krebs et al., 2018; Oli et al., 2022). Data were insufficient to perform more detailed, cycle-specific analyses for all five cycles, which is a limitation of our paper. Also, it is challenging to quantify the reproductive performance of females in the field, and we require either shot samples (Cary & Keith, 1978; Keith & Windberg, 1978) or capture and rearing of the pregnant females in maternity cages just before they give birth (Hodges et al., 2001; Stefan, 1998; Stefan & Krebs, 2001). Both approaches are invasive and require substantial time and effort. Whereas our estimates of survival and transition probabilities are based on CMR data collected over five cycles (1977–2020), litter size and pre-weaning survival data were available only for one cycle (1989–1996). Our estimates of survival rates were apparent, rather than true, survival rates because loss included both death and dispersal; consequently, survival rates of the dispersing age class, particularly of juveniles, were underestimated. Although pre-weaning survival rates were estimated based on radio-telemetry data and represent true survival, Stefan (1998) and Hodges et al. (2001) acknowledge the possibility of underestimation of this parameter as well. Thus, it is possible that λ may have been slightly underestimated. Despite these potential limitations, we believe that our findings are robust and general enough to capture the seasonal and phase-specific variation in demography and population dynamics of snowshoe hares.

5 | CONCLUSIONS

Consistent with our predictions, (1) adult hares had a higher survival probability than juveniles across all seasons and phases; (2) state-specific survival exhibited strongly phase-specific patterns with the lowest survival during the decline phase compared with other cyclic phases; (3) hares of both sexes and age classes exhibited strongly seasonal patterns with lower survival during winter than during summer months, except that overwinter survival of juveniles was fairly high, likely because of the fact that first and second litter juveniles would have grown to be similar in size to adults by autumn; and (4) the probability that non-reproductive females become reproductive, and reproductive females remaining reproductive was higher during the increase/peak phase than other cyclic phases. However, contrary to our expectations and predictions of the life-history theory for species with fast life histories (Hamel, Gaillard, & Yoccoz, 2010; Hamel, Gaillard, Yoccoz, Loison, et al., 2010), there was no evidence for the cost of reproduction in our study population because reproductive females survived as well as, or better than, non-reproductive adult females or adult males. The lack of evidence for the survival cost of reproduction may be a result of heterogeneity in individual quality; high quality females may be able to successfully reproduce but also have high survival due to good physical condition and high resource acquisition efficiency (Cooch et al., 2002; Hamel, Gaillard, & Yoccoz, 2010; van Noordwijk & de Jong, 1986), and possibly due to a higher abundance of alternative prey during summer, the primary breeding season. Finally, consistent with our expectations, (5) the population growth rate, generation time and other population-dynamic quantities exhibited strong seasonal and phase-related patterns.

Although the mechanistic causes of population cycles in snowshoe hares and arvicoline rodents continue to be debated (Edwards et al., 2023, 2025; Krebs, 2013; Krebs et al., 2018; Myers, 2018; Oli, 2019), our understanding of demographic processes underlying snowshoe hare population cycles is improving (Boonstra, Krebs, & Stenseth, 1998; Cary & Keith, 1978; Hodges et al., 1999, 2001; Krebs et al., 1986; Oli et al., 2020). Through the application of the state-of-the-science CMR models to unprecedented field data spanning five population cycles, we provide a comprehensive assessment of state-specific demographic patterns, and the complex manner in which cyclic phases interact with seasonality to influence these patterns. For example, survival of juveniles is surmised to be potentially important in driving snowshoe hare population cycles (Hodges et al., 1999). Here, we show that the probability of juveniles surviving from weaning until the following spring is three times higher during the increase phase than the decline phase. Previous studies have shown that hares survive better in summer than in winter (Krebs et al., 1986; Oli et al., 2020, 2022). Here, we show that phase-specific survival of both sexes and age classes changes as the reproductive season progresses. We show that reproductive transition probabilities as well as the proportion of females that were reproductive vary strongly across seasons and phases, and so do pre-weaning survival and litter

sizes. Finally, we combined estimates of survival and reproductive transition probabilities with data-based estimates of reproductive parameters to construct and analyse season- and phase-specific patterns of demography and population dynamics. In addition to season and phase-related differences in asymptotic λ , we show that snowshoe hares exhibit some life-history and demographic characteristics that are expected for species with 'fast' life history, and sometimes those expected of species with slow life histories. It is remarkable that the generation time can vary by 2.4-fold, RVs of adult females by fivefold and life expectancy by fourfold, depending on the season and phase of the population cycle. We are not aware of these levels of life-history and population-dynamic variation within the same population. Previous studies have surmised that juvenile survival may play an important role in driving snowshoe hare population cycles (Keith & Windberg, 1978; Peers et al., 2021); here, we unambiguously establish the prominent role of pre-weaning survival in driving snowshoe hare population cycles. Despite these novel insights provided by this study, our demographic analyses have some inherent limitations. For example, all the demographic quantities derived from matrix population models are asymptotic (and not observed) values calculated under the assumption of the stable age/stage theory (Caswell, 2001), which may not strictly hold in cyclic populations; in such cases, analysis of transient dynamics would be very informative (Caswell, 2007; Dobson et al., 2024; Mertens et al., 2006). Also, demographic parameters in our study population exhibit strongly seasonal patterns, and theory exists for the analysis of seasonal matrices (e.g. Caswell, 2001; Caswell & Trevisan, 1994). However, the current theory of seasonal matrix models is difficult, if not impossible, to apply to our study system because season and phase are confounded—a given season can be in any of the four cyclic phases (increase, peak, decline or low). We also note that our primary goal for these matrix model analyses was to characterize phase and season-specific demographic patterns, and not to mimic the dynamics of our study population. Future modelling efforts could develop and apply the theory of transient and seasonal dynamics in cyclic populations where vital rates are not only seasonal but also phase-specific. Whereas phase-related variation in survival is most likely a consequence of predator abundance (Krebs et al., 1995, 2018), causes of phase-specific variation in litter size, pre-weaning survival and the probability of non-reproductive females becoming reproductive, and reproductive females remaining reproductive across all seasons are yet to be ascertained. Predation of litters prior to weaning, predator-mediated changes in foraging behaviour leading to poorer female body condition and/or reduced parental care (Boudreau et al., 2019; Majchrzak et al., 2022), and maternally inherited predator-induced stress have been proposed as a possible explanation for the phase-related transition probabilities we observed (Boonstra, Hik, et al., 1998; Boonstra, Krebs, & Stenseth, 1998; Sheriff et al., 2010, 2011, 2015, 2017), but evidence is as yet inconclusive. Findings of earlier studies (Cycles 2 and 4) were consistent with the predictions of the stress hypothesis (Boonstra, Hik, et al., 1998; Boonstra, Krebs, & Stenseth, 1998; Sheriff et al., 2011), but a recent study in Cycle 5 by Lavergne et al. (2021) failed to find a

physiological signature indicative of chronic predator-induced stress reported by the previous studies. Winter food shortages have also been hypothesized to cause reproductive changes in snowshoe hares in Alberta (Keith, 1983; Keith & Windberg, 1978), but earlier experimental evidence did not support this hypothesis (Krebs et al., 1995, 2018). A recent individual-based food supplementation experiment found that food supplementation doubled overwinter survival, and that supplemented hares lost less body mass than control hares (Majchrzak et al., 2022). It is, therefore, possible that supplemented hares would be in better condition and thus would have higher reproductive potential than control hares. In a 16-year study, Sinclair et al. (2003) showed that female hares collected from high density phase had lower reproductive rates relative to those collected from the low phase of the cycle; the reduced reproductive rates led to the extinction of high-phase lineages. These findings, along with similar results from cyclic rodent populations (Boonstra, 2024; Edwards et al., 2021; Mihok & Boonstra, 1992) point to the possibility of differences among females in alternating cyclic phases, which may work synergistically with environmental causes like predation to cause population cycles. Whereas the Kluane experiment (Krebs et al., 1995; Krebs, Boutin, & Boonstra, 2001) has provided strong evidence to support the idea that delayed density-dependent predation is the primary cause of phase-related changes in snowshoe hare survival, the mechanisms that permit high reproductive transitions, larger litters and improved pre-weaning survival during the increase phase are less well understood (Boudreau et al., 2019; Lavergne et al., 2021). Our study represents the first comprehensive demographic analysis of any cyclic population based on long-term field data and state-of-the-science modelling techniques and provides a thorough description of the demographic mechanisms underlying snowshoe population cycles. We suspect that our results are general and apply to many other cyclic mammalian populations; similar studies of other cyclic mammals would be needed to test the generality of our findings.

AUTHOR CONTRIBUTIONS

C.J.K.: Conceptualization; data curation; funding acquisition; investigation; methodology; project administration; supervision; writing—original draft; writing—review and editing. A.J.K.: Data curation; project administration; resources; writing—review and editing. R.B. and S.B.: Funding acquisition; investigation; methodology; project administration; writing—review and editing. D.L.M.: Investigation; methodology; writing—review and editing. T.S.J.: Data curation; investigation; writing—review and editing. J.E.H.: Formal analysis; software; visualization; writing—review and editing. M.K.O.: Conceptualization; data curation; formal analysis; investigation; methodology; visualization; writing—original draft; writing—review and editing. All authors gave final approval for publication of the manuscript.

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CONFLICT OF INTEREST STATEMENT

All authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data are available from Figshare Repository. <https://figshare.com/s/3dcca1530bc99f163772>. DOI: [10.6084/m9.figshare.28001726](https://doi.org/10.6084/m9.figshare.28001726) (Oli et al., 2025).

ETHICS STATEMENT

Animal research protocol was approved by the University of British Columbia (Animal Care Committee, certificate A17-0111), which adheres to the guidelines of the Canadian Council on Animal Care.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Table S1. Global goodness-of-fit-test (GOF) results for the multistate capture–mark–recapture model performed with U-Care software.

Figure S1. (A) The number of snowshoe hares captured each month by sex and age classes. Months are numbered sequentially from 2 (February) to 12 (December); and (B) proportion ($\pm 95\%$ CI) of captured adult females that were reproductive during each cyclic phase.

Figure S2. State-specific survival (A) and transition (B) probabilities ($\pm 95\%$ CI) for snowshoe hares at Kluane, Yukon, Canada, 1977–2020, based on a multistate capture–mark–recapture model that does not include cyclic phase or season.

Figure S3. (A) State- and phase-specific survival, and (B) transition probabilities ($\pm 95\%$ CI) for snowshoe hares at Kluane, Yukon, Canada, 1977–2020, based on a multistate capture–mark–recapture model that includes cyclic phase but not season.

Figure S4. State- and season-specific survival (A) and transition probabilities (B) ($\pm 95\%$ CI) for snowshoe hares at Kluane, Yukon, Canada, 1977–2020.

Figure S5. State, phase, reproductive season and cycle number specific estimates of capture probabilities (p ; vertical lines indicate 95% confidence intervals) for snowshoe hares at Kluane, Yukon, Canada, 1977–2020.

Figure S6. Reproductive season- and phase-specific estimates of pre-weaning survival per month (A) and litter size (B) in a cyclic population of snowshoe hares at Kluane, Yukon, Canada, 1989–1996.

Figure S7. Sensitivity of finite population growth rate (λ) to changes in pre-weaning survival (s_0), survival of juveniles (s_1), survival of non-reproductive adult females (s_2), survival of reproductive adult females (s_3), conditional transition probability of juveniles to become nonreproductive (ψ_{fn} ; Ψ_{fn}) and reproductive (ψ_{fr} ; Ψ_{fr}) adults, conditional transition probability of non-reproductive adult females to become reproductive (ψ_{nr} ; Ψ_{nr}), conditional transition probability of reproductive females to become non-reproductive (ψ_{rn} ; Ψ_{rn}) and conditional transition probability of reproductive female to remain reproductive (ψ_{rr} ; Ψ_{rr}).

Figure S8. Differences in asymptotic monthly finite population growth rate between cyclic phases ($\Delta\lambda$) during each reproductive season in a cyclic population of snowshoe hares at Kluane, Yukon, Canada.

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